

THANUSRI AENUGULA

+1 940 364 1654 | thanusri1601@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio Website](#)

Education

Texas A&M University, College Station

Aug 2025 - Dec 2026

Master of Science in Data Science

CGPA: 3.75/4

Relevant Courses: Data Mining, Statistics, Mathematics for ML, Machine Learning, Deep Learning, Reinforcement Learning

Sri Sivasubramaniya Nadar College of Engineering

Jul 2021 - Jun 2025

Bachelor of Engineering in Electrical and Electronics Engineering

CGPA: 3.8/4

Relevant Courses: Data Structures, Probability and Statistics, Database Management, Data Modeling, NLP, Artificial Intelligence

Technical Skills

Programming Languages: Python, SQL, R, C++, Matlab, HTML, CSS, JavaScript

Libraries & Frameworks: PyTorch, TensorFlow, Transformers, LangChain, LangGraph, NumPy, Pandas, OpenCV, NLTK, PySpark

Tools: AWS (SageMaker Studio, Bedrock, S3, EC2, Lambda), Streamlit, Git, MySQL, MongoDB

AI/ML Architectures: CNN, RNN, LSTM, Autoencoder, LLM, Supervised Learning, Clustering

Experience

Texas A&M University | *Research Assistant, Advance Vision and Learning Lab*

Dec 2025 - Present

- Establishing centralized data pipelines to integrate **3+ modalities** and analyze data for automated phenotyping across multiple tasks.
- Supporting analysis of **5+ plant growth traits** using AI and statistical modeling techniques enabling automated phenotyping.

Texas A&M University | *Research Volunteer, Department of Animal Science*

Aug 2025 - Dec 2025

- Designed multi-view feature fusion pipelines that boosted behavioral classification accuracy by **27%** using spatio-temporal camera cues.
- Executed real-time video analytics and data streaming workflows to process **200+** hours of barn footage weekly for monitoring.
- Refined **temporal attention networks** to detect subtle behavioral transitions, allowing earlier identification of feeding anomalies.

Daimler India Commercial Vehicles | *Data Science Intern*

Jan 2025 - Jun 2025

- Collaborated with the Data-Driven Engineering team to develop machine learning tools for the **cross-functional team**, streamlining **5+** vehicle testing workflows.
- Built a **statistical twin** for Dynamic Brake Testing systems using historical brake from **100+** vehicles enriching predictive analysis.
- Delivered a production-ready solution that enabled testing team to pre-screen vehicle performance, reducing testing time by **80%**.
- Investigated telematics and Dealer Management Service data to build **ETL pipelines** for real-time monitoring, supporting driving pattern analysis and fleet optimization using **Azure Databricks**.

Defence Research and Development Organisation (DRDO) | *Data Science Intern*

Sep 2024 - Nov 2024

- Integrated **CNN**, **LSTM**, and **XGBoost** to predict battery Remaining Useful Life and State of Charge, achieving **R² of 0.97** and strengthening forecasting accuracy.
- Examined performance data from **38+** lithium-ion batteries, reducing failure prediction time by **25%** and identifying early degradation.
- Architected monitoring pipelines to track battery health helping the team plan maintenance and manage assets more effectively.

Projects

Multi-Agent SQL Query Execution System [[Github](#)]

- Designed a LangGraph-based **multi-agent** system automating SQL analytics and reporting, reducing manual effort by **60%**.
- Built a SQL reasoning pipeline with enforced grounding and summarization constraints, ensuring **100%** query execution validity.
- Deployed on **AWS EC2**, enabling real-time multi-agent inference and **sub-2s** downloadable PDF report generation.

Corrective RAG System for AI Knowledge Retrieval [[Github](#)]

- Designed a system using **LangGraph** to validate retrieval context and correct queries, reducing unsupported LLM responses by **30%**.
- Built an end-to-end RAG pipeline with web scraping, semantic chunking, and **FAISS** vector search for low-latency, grounded retrieval.
- Deployed a Streamlit application on **AWS EC2**, enabling real-time inference with **low response latency**.

LLM Adaptation for Financial Text-Classification [[GitHub](#)]

- Fine-tuned a **RoBERTa** model with **LoRA (PEFT)**, reducing trainable parameters and GPU memory usage by **70-80%**.
- Built an end-to-end NLP pipeline for 3-class sentiment classification, covering data preprocessing, tokenization, training, and evaluation.
- Deployed the fine-tuned model on the **Hugging Face Hub**, enabling real-time sentiment inference via a public interface.

Scalable NLP Classification System [[GitHub](#)]

- Created scalable text classification system on a Wikipedia-derived dataset, achieving **98.68% accuracy** using **TF-IDF + Linear SVM**.
- Fine-tuned a **DistilBERT** transformer with **AdamW** optimization, enhancing performance to **99.27% accuracy** across all classes.
- Applied **LIME** interpretability to compare model decisions, showing lexical reliance in SVM versus understanding in DistilBERT.

Predictive Maintenance of Wind Turbine Systems [[GitHub](#)]

- Modeled hybrid deep learning models for Remaining Useful Life prediction of wind turbine systems, with a **CNN-LSTM-GRU** model achieving an **R² of 0.91**.
- Crafted an interactive **Power BI** dashboard to monitor **10+ KPIs** and visualize system performance trends.

Behavioural Churn Modeling [[GitHub](#)]

- Improved churn prediction accuracy to **98%** using **Gradient Boosting**, facilitated retention strategies, and reduced churn by **15%**.
- Optimized accuracy by conducting 30+ experiments with diverse hyperparameters, refined feature engineering, reducing **false positives**.
- Validated model stability using **cross-validation**, achieving **< 10%** difference between training and testing metrics.